

**CSE260 Assignment 01**  
**[C01]**  
**Total Mark: 50**  
**Deadline: July 9, 2025; 11:59PM**  
**Submission Link: [Click Here](#)**

**This assignment must be hand-written. Show ALL steps in ALL questions.**

1. Convert the following binary numbers to equivalent decimal numbers.  
(a)  $(101110001001)_2$   
(b)  $(11011.101)_2$
2. Convert the following decimal number to equivalent base 5 numbers:  $(4123)_{10}$
3. Convert the following decimal number to equivalent binary numbers:  $(4785.150263)_{10}$  [for infinite fractional part, just do 6-7 steps and use dots for the rest]
4. Perform the following base conversions  
(a)  $(29)_{12} = (?)_7$   
(b)  $(10110111)_5 = (?)_4$
5. Perform addition, subtraction and multiplication for the pair of following base-9 numbers. Verify your results by converting the problem into decimal.  
 $(412, 134)$
6. Convert the following decimal number to Excess-3 showing calculation:  
 $(745.42)_{10}$
7. Subtract 45 from 98 in 12 bits using 2's complement number system and justify whether there is an overflow or not.
8. Add 13 with 27 in 6 bits using 2's complement number system and justify whether there is an overflow or not.
9. Add -57 with 63 in 11 bits using 1's complement number system and justify whether there is an overflow or not.
10. You are a computer engineer and you want to buy two 8 GB DDR4 RAMs. Each RAM costs  $(1C2)_{16}$  dollars. You also want to buy a graphics card RTX4070Ti

which costs  $(10010110000)_2$  dollars. However, you don't have that much money with you and you are afraid to ask your parents about it. Suddenly, one of your generous friends agreed to give you the money you need. He decided to give you  $(4064)_8$  dollars. How much will you have left after buying those components? (Show the answer in decimal)